Putting it all together - Work-Energy Theorem.

For each example below, fill in the table (overpage) with the required quantities

A A man lifts a 5 kg rucksack from the floor to a hook, 2m above the ground. The rucksack starts and ends at zero speed. ***System = only the rucksack***

B Same problem but ***system = rucksack + earth***

C A girl drags a 10 kg sandbag along the horizontal floor for a distance of 2.5 m applying a constant horizontal pulling force of 8N on the sandbag. Friction between sandbag and ground is assumed to be negligible. Sandbag initially stationary. ***System = the sandbag***.

D The girl drags the 10 kg sandbag along the horizontal floor for a distance of 2.5 m applying a constant horizontal pulling force of 8N on the sandbag, maintaining the sandbag at a constant speed of 1.0 m/s. ***System = the sandbag.***

E The girl drags the 10 kg sandbag along the horizontal floor for a distance of 2.5 m applying a constant horizontal pulling force of 8N on the sandbag. Average friction is assumed to be 2N. Sandbag initially stationary. ***System = the sandbag.***

F The girl drags the 10 kg sandbag up an inclined plane, increasing its height by 10 cm, for a distance of 2.5 m applying a constant horizontal pulling force of 8N. Average friction is assumed to be 2N. Sandbag initially moving at 1.0 m/s. ***System = the sandbag + Earth.***

G The girl drags a 10 kg sandbag up an inclined plane, increasing its height by 10 cm, for a distance of 2.5 m applying a constant horizontal pulling force of 8N. Average friction is assumed to be 2N. Sandbag initially moving at 1.0 m/s. ***System = the sandbag.***

H A 65 kg box is pushed up a frictionless slope inclined at 30° to the horizontal, applying a force of 600 N (parallel to the surface), over a distance of 2 m. ***System = the box***

I Same problem, but ***system = the box + the Earth***

J A 65 kg box is pushed up a slope inclined at 30° to the horizontal, applying a force of 600 N (parallel to the surface), over a distance of 2 m. The average frictional force is 5 N . ***System = the box***

K Same problem, but ***system = the box + the Earth***

L A spring with spring constant 2.5N/cm is preloaded by compressing to half its normal length (which is 8.0 cm) It is used to fire a ball, mass 0.1 kg. ***System = ball***

M Same problem, but ***system = ball + spring***

N A spring mounted lever on a pin-ball machine is used to accelerate a ball as it is ascending a slope. The ball, mass 45 g, is moving up the slope at a speed of 0.64 m/s when the spring mounted lever releases, accelerating the ball up the slope. A certain time later the ball has increased in height by 8.5 cm and is moving at a speed of 1.2 m/s. During this time the ball has moved a distance of 55 cm and has experienced an average frictional force of 0.14 N. ***Take the system to be the earth, the ball and the spring***

O Same problem, but take the ***system to be only the ball.***

The table shows the quantities as described in the work-energy theorem

(if it helps, for Ug, put (Ug)i = 0, then (Ug)f = ΔUg

	All in joules						
Situ	K_i	U_{gi}	U_{si}	W	K_f	U_{gf}	U_{sf}
A							
B							
C							
D							
E							
F							
G							
H							
I							
J							
K							
L							
M							
N							
O							

Notes:

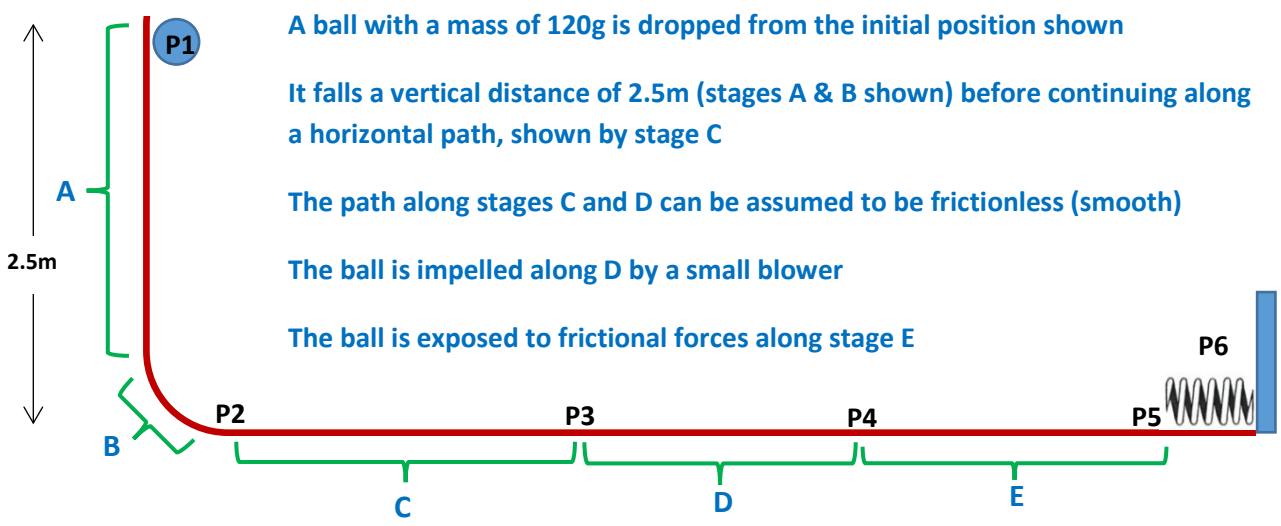
The origin of energy to provide positive external work can be from a variety of sources (e.g. chemical, electrical, etc.)

When we have negative work contributions, this implies that energy is leaving the system and therefore increasing somewhere else. For example: the height of an object in a gravitational field or the internal energy (production of heat).

The work-energy equation

Unless stated, assume frictional forces and air resistance are negligible. Take g = 10 N/kg

The system consists of the spring, the blue ball and the Earth.



The ball is released from the position shown, P1.

It travels down the track until it is moving horizontally along stage C (from P2 to P3)

Once it reaches stage D a small electric blower is used to apply a constant force of 2.2N to the right, and the force is applied until the ball reaches stage E, 1.2m later (from P3 to P4)

Once the ball reaches stage E it is now exposed to a constant frictional force of 2.6N for 1.4m (P4 to P5) until it collides with the spring coming to a temporary halt when the spring is fully compressed (P5 to P6)

1. Identify and discuss each energy transfer taking place from P1 → P2 → P3 → P4 → P5 → P6

2. Write down the equation comparing the energy states of the system initially (at P1) and finally (at P6). Include in your equation the work done on the system. Determine each term in your equation.

3. Assuming there is no transfer to internal energy during the subsequent decompression of the spring, calculate the initial speed of the ball when it is projected.

4. Given that the spring had shortened by 2.5cm (P5 / P6) calculate its stiffness constant.

5. Considering only the ball and the Earth as the system, reason whether the work done by the following forces on the ball is positive, negative or zero:

   (a) Gravity during stage A

   (b) The slope, during stage B

   (c) The air force, during stage D

   (d) The surface, during stage E

Summary:

For any system:

The initial mechanical energy of the system + the work done on the system equates to the final mechanical energy of the system

Work-Energy Theorem:

If an object is taken to be the system, the net work done on an object is equal to the gain in kinetic energy

We can state this more simply by saying: the net work done on an object is equal to the gain in kinetic energy of the object.

This is often referred to as the work-energy theorem